

Anti-Theft

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

The Body Control Module (BCM) controls the operation of the perimeter alarm based on inputs from all the door ajar switch inputs, hood switch input, keyless entry keypad, door lock control switches, Passive Anti-Theft System (PATS) input, and the Remote Keyless Entry (RKE) transmitters. If the perimeter alarm is armed, and the BCM receives an input indicating an unauthorized entry into the vehicle or the ignition is turned to the RUN position without a programmed PATS key, the BCM activates the perimeter alarm. When the perimeter alarm is activated, the horn sounds and both the turn signals and courtesy lights flash in regular intervals.

If the driver door is unlocked using a key in the door lock cylinder and then opened, the perimeter alarm starts a 12-second countdown. During the 12-second countdown there is a chime from the Instrument Panel Cluster (IPC) and the message center displays TO STOP ALARM START VEHICLE. If the alarm is not deactivated during this countdown, the perimeter alarm activates.

Battery Disconnect

If the battery is disconnected and then reconnected, the perimeter alarm remains in the same state (disarmed, pre-armed, armed, or activated) it was in prior to the battery disconnect.

Alarm Event PIDs

The BCM has PIDs available that show what caused the last alarm events. The 4 BCM alarm event PIDs are ALARM_EVT_1 through 4. The cause for the most recent alarm activation is always listed in the first alarm event PID. These PIDs display what caused the alarm to activate and can be a very useful tool when trying to identify the cause of a false alarm activation.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of electrical damage.

Visual Inspection Chart

Electrical
• Wiring, terminals or connectors • Hood switch • Body Control Module (BCM)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00 to diagnose No Power To The Scan Tool.

6. If the scan tool does not communicate with the vehicle:

- verify the ignition key is in the ON position.
- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00 to diagnose No Response From The PCM.

7. Carry out the network test:

- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
- If the network test passes, retrieve and record the continuous memory DTCs.

8. Clear the continuous DTCs and carry out the self-test diagnostics for the BCM .

9. If the DTCs retrieved are related to the concern, go to the DTC Chart. For all other DTCs, refer to the DTC Chart in Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

Body Control Module (BCM) Diagnostic Trouble Code (DTC) Chart

BCM DTC Chart

DTC	Description	Action
B1305:01	Hood Switch: General Electrical Failure	GO to Pinpoint Test B.
All other DTCs	—	REFER to the Master DTC Chart in Section 419-10.

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
The alarm system does not activate from an unauthorized entry at the hood	- Wiring, terminals or connectors - Hood switch - Body Control Module (BCM)	GO to Pinpoint Test A.
The alarm system does not activate from an unauthorized entry at a door	Door ajar input	REFER to Section 417-02.
The alarm system does not arm/no turn signal flash confirmation	- Door ajar input - Hood ajar input - Turn signals - Keyless entry keypad concern - Remote Keyless Entry (RKE) transmitter concern - BCM	GO to Pinpoint Test C.
No horn output during alarm activation	Horn	REFER to Section 413-06.
No turn signal flash during alarm activation	Turn signals	REFER to Section 417-01.
No courtesy lamp flash during alarm activation	Courtesy lamps	REFER to Section 417-02.
False alarm activation	- Door ajar input - Hood switch input - Panic button pressed on an IKT - A valid Passive Anti-Theft System (PATS) key read is NOT received when the ignition is turned to RUN	An intermittent fault exists. MONITOR the BCM alarm event PIDs ALARM_EVT_1 through 4 to identify and isolate the cause of the last alarm activations.

• All keys/RKE transmitters lost, perimeter alarm activated, and unable to program new keys	• Customer lost all keys/RKE transmitters	• PROGRAM an RKE transmitter. REFER to Section 501-14. UNLOCK the doors with the programmed RKE transmitter. PROGRAM the new keys. REFER to Section 419-01B.
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Pinpoint Tests

Pinpoint Test A: The Alarm System Does Not Activate From An Unauthorized Entry At The Hood

Refer to Wiring Diagrams Cell [117](#) , Remote Keyless Entry and Alarm for schematic and connector information.

Normal Operation

When the alarm system is armed, the Body Control Module (BCM) monitors the hood switch. If the hood is opened while the alarm is armed, the hood switch removes ground from the signal circuit. This signals the BCM that an unauthorized intrusion to the engine compartment has occurred and the BCM activates the alarm.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Hood switch
- BCM

PINPOINT TEST A : THE ALARM SYSTEM DOES NOT ACTIVATE FROM AN UNAUTHORIZED ENTRY AT THE HOOD

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

A1 RETRIEVE THE RECORDED DTCS IN THE BCM

NOTE: DTC B1305:01 sets if the hood is open. Make sure the hood is closed prior to running the self-test.

- Retrieve the recorded DTCs from the Body Control Module (BCM) self-test.

Is DTC B1305:01 present?

Yes	GO to Pinpoint Test B.
No	GO to A2 .

A2 CHECK THE HOOD SWITCH (HOOD_SW) PID

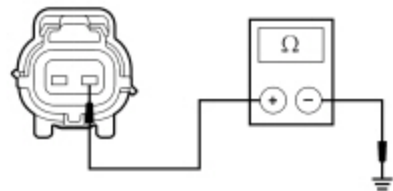
- Enter the following diagnostic mode on the scan tool: BCM DataLogger .
- Disconnect: Hood Switch [C127](#).
- Monitor the BCM PID HOOD_SW while disconnecting the hood switch.

Does the BCM PID read AJAR when the switch is disconnected?

Yes	INSTALL a new hood switch. TEST the system for normal operation.
No	GO to A3 .

A3 CHECK THE HOOD SWITCH INPUT CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: BCM [C2280F](#) .
- Measure the resistance between the hood switch [C127](#) Pin 1, circuit CPL25 (BU/OG), harness side and ground.



• N0056246

Is the resistance greater than 10,000 ohms?

Yes	GO to A4 .
No	REPAIR circuit CPL25 (BU/OG) for a short to ground. TEST the system for normal operation.

A4 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test B: DTC B1305:01

Refer to Wiring Diagrams Cell [117](#) , Remote Keyless Entry and Alarm for schematic and connector information.

When the alarm system is armed, the Body Control Module (BCM) monitors the hood switch. If the hood is opened while the alarm is armed, the hood switch removes ground from the signal circuit. This signals the BCM that an unauthorized intrusion to the engine compartment has occurred and the BCM activates the perimeter alarm.

- DTC B1305:01 (Hood Switch: General Electrical Failure) — set by the BCM if it detects an open on the hood switch sense circuit. This DTC also sets if the hood is open during a BCM self-test.

This pinpoint test is intended to diagnose the following:

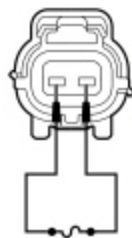
- Wiring, terminals or connectors
- Hood switch
- BCM

PINPOINT TEST B : DTC B1305:01

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

B1 CHECK THE HOOD SWITCH

- Disconnect: Hood Switch C127.
- Connect a fused jumper wire between the hood switch C127 Pin 1, circuit CPL25 (BU/OG), harness side and the hood switch C127 Pin 2, circuit GD120 (BK/GN), harness side.



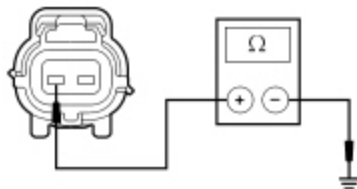
- N0112495
- Clear the BCM DTCs.
- Repeat the BCM self-test.

Is the BCM DTC B1305:01 retrieved?

Yes	REMOVE the jumper wire. GO to B2.
No	REMOVE the jumper wire. INSTALL a new hood switch. TEST the system for normal operation.

B2 CHECK THE HOOD SWITCH GROUND CIRCUIT

- Measure the resistance between the hood switch C127 Pin 2, circuit GD120 (BK/GN), harness side and ground.



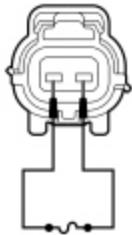
- N0082085

Is the resistance less than 5 ohms?

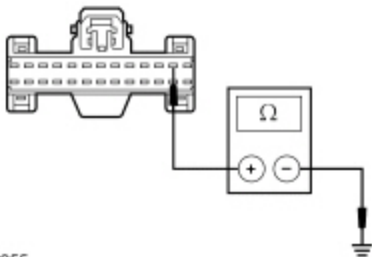
Yes	GO to B3 .
No	REPAIR circuit GD120 (BK/GN) for an open. TEST the system for normal operation.

B3 CHECK THE HOOD SWITCH SIGNAL CIRCUIT TO THE BCM FOR AN OPEN

- Disconnect: BCM [C2280F](#) .
- Connect a fused jumper wire between the hood switch [C127](#) Pin 1, circuit CPL25 (BU/OG), harness side and the hood switch [C127](#) Pin 2, circuit GD120 (BK/GN), harness side.



- N0112495
- Measure the resistance between the BCM [C2280F](#) Pin 2, circuit CPL25 (BU/OG), harness side and ground.



• N0090255

Is the resistance less than 5 ohms?

Yes	GO to B4 .
No	REPAIR circuit CPL25 (BU/OG) for an open. TEST the system for normal operation.

B4 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test C: The Alarm System Does Not Arm/No Turn Signal Flash Confirmation

Normal Operation

The Body Control Module (BCM) controls the perimeter alarm system and based on inputs either arms, disarms or activates the alarm. When the alarm is activated, the BCM sounds the horn and flashes both the turn signals and courtesy lamps in regular intervals. The BCM receives inputs for the door lock status, hood ajar, door ajar and the keyless entry keypad. The BCM also receives signals from the Remote Keyless Entry (RKE) transmitters.

For the perimeter alarm to arm and provide a flash of the turn signals, the following conditions must be met:

- Ignition must be OFF
- The hood closed
- All doors closed
- An electronic lock command received from an RKE transmitter, keyless entry keypad, or door lock control switch (front door opened and then shut after lock is pressed).

Once all the conditions are met, the alarm system enters a pre-arm phase and begins a 20-second countdown. After 20 seconds have passed and no unlock command is received or any hood or door is opened, the perimeter alarm arms.

The perimeter alarm is activated when:

- any passenger door or the hood is opened without an electronic unlock command from an RKE transmitter or keyless entry keypad received first.
- the driver door is opened without an unlock command from an RKE transmitter or keyless entry keypad received first and the 12 second countdown has expired.
- the ignition is turned to the RUN position without a valid Passive Anti-Theft System (PATS) key read received.

This pinpoint test is intended to diagnose the following:

- Door ajar input
- Hood switch input
- Turn signals
- Keyless entry keypad concern
- RKE transmitter concern
- BCM

PINPOINT TEST C : THE ALARM SYSTEM DOES NOT ARM/NO TURN SIGNAL FLASH
CONFIRMATION

C1 RETRIEVE THE RECORDED DTCS IN THE BCM

- Retrieve the recorded DTCs from the Body Control Module (BCM) self-test.

Is DTC B1305:01 present?

Yes	GO to Pinpoint Test B.
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No	GO to C2 .
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C2 CHECK THE COURTESY LAMP OPERATION

- Open and close all the doors while checking the courtesy lamp operation.

Do the courtesy lamps operate correctly?

Yes	GO to C3 .
No	REFER to Section 417-02 to diagnose the door ajar inputs.

C3 VERIFY THE HAZARD LAMP OPERATION

- Place the hazard switch in the ON position.

Do the hazard lamps operate correctly?

Yes	GO to C4 .
No	REFER to Section 417-01 to diagnose the hazard lamps.

C4 CHECK THE RKE TRANSMITTER FUNCTION

- Using the ~~RKE~~ transmitter, lock and unlock the doors and operate the panic function.

Do the ~~RKE~~ transmitter functions operate correctly?

Yes	GO to C5 .
No	REFER to Section 501-14 to diagnose the RKE function.

C5 CHECK THE KEYLESS ENTRY KEYPAD FUNCTION

- Lock the doors using the keyless entry keypad.

Do all the doors lock using the keyless entry keypad?

Yes	GO to C6 .
No	REFER to Section 501-14 to diagnose the keyless entry keypad.

C6 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the ~~BCM~~ connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins

- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.